

Miguel Contreras

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EDUCATION

University of California, Berkeley

Berkeley, CA

Bachelor of Arts in Computer Science, Data Science, Education

Expected May 2027

Coursework: Machine Learning; Data Science; Computer Architecture; Data Structures and Algorithms; Probability

EXPERIENCE

Software Engineer Intern

May 2026 – Aug 2026

Amazon

Redmond, WA

- Architected and delivered an E2E SDK observability platform spanning a CloudWatch data-ingestion pipeline, AWS Lambda, and React, giving 8+ teams real-time visibility into SDK health across all external partners.
- Built a full-stack internal dashboard in React (Harmony) with Midway/SSO auth, backed by an AWS Lambda BFF service, serving as the single source of truth for engineers monitoring SDK customer health.
- Built a production pipeline ingesting gateway request logs into CloudWatch, producing real-time views of API call volume, error attribution, and SDK version adoption grouped by customer account and operation.

Software Engineer Contract

Jan 2026 – May 2026

Google

Mountain View, CA

- Evaluated Google's GenAI stack (AI Studio, AlloyDB, Cloud Run) by building a full-stack app to surface developer friction, filing issues to Google's source repos and presenting findings to engineering teams.

Software Engineer Intern

May 2025 – Aug 2025

Amazon

Redmond, WA

- Architected and delivered a full-stack data retention query management system that transformed manual 48–72 hour investigative workflows into automated processes completing in 5–10 minutes, serving global customers.
- Deployed and fine-tuned LLM-powered applications on AWS Bedrock to translate natural language into optimized SQL queries, using retrieval-augmented generation (RAG) for search and data access.
- Integrated 4+ AWS services (Athena, S3, Bedrock, GLUE) into a secure, production-ready architecture with real-time query monitoring, automated file handling, cross-account authentication, and compliance with international data protection regulations.

Data Engineer

Jan 2025 – May 2025

Bay FC — National Women's Soccer League

San Jose, CA

- Designed and deployed personalization and user modeling algorithms by applying KNN, PCA, and KMeans clustering to classify players into tactical archetypes, analogous to recommendation system personalization.
- Implemented dimensionality reduction (PCA) and KMeans clustering to identify and visualize player roles.
- Engineered player-level features (off-ball movement, sprint patterns, line-breaking actions) to predict user engagement-like metrics, aligning with CTR/CVR-style prediction and ranking tasks.

PERSONAL PROJECTS

Refill! – *Typescript, Google AlloyDB, Gemini API, React, PostgreSQL, Docker*

- Designed, built, and deployed a production social app that lets users find, submit, and review water refill stations on an interactive Google Maps interface, with Google OAuth sign-in, photo uploads, a personalized activity feed, mutual friend graph, and refill leaderboard. Powered by Google AlloyDB and shipped Google Cloud Run.

Acne Detection & Cross-Domain Classification – *Python, PyTorch, YOLOv5*

- Trained and benchmarked 3 object detection architectures (YOLOv5, Faster R-CNN, DINO-DETR) on the ACNE04 dataset (1,450 images) to localize acne lesions, evaluating models on mAP, precision/recall, and IoU.
- Engineered a cross-domain classification pipeline by extracting positive/negative patches from bounding box annotations and training a binary classifier to generalize from ACNE04 to the DermNet dataset

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, RISC-V, Rust, Scheme

Frameworks: Pytorch, TensorFlow, Tailwind, React, Node.js, FastAPI, OpenCV, Pandas, Numpy

Developer Tools: Git, Google Cloud, Linux, AWS, IAM, Databases, Glue, S3 Buckets, Athena